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Multimodal artificial intelligence for human–computer interaction enhanced breast lesion diagnosis

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ABSTRACT

This study aims to enhance breast ultrasound diagnosis by developing a multimodal artificial intelligence (AI) system that integrates lesion detection, diagnostic fusion, and human–computer interaction. We propose KIDNet, a deep learning model for detecting and scoring lesions. A Transformer-based network then fuses ultrasound images, AI-derived scores, and clinical data for final diagnosis. Our results demonstrate that the multimodal fusion approach achieves a diagnostic accuracy of 93.1%, significantly outperforming unimodal and simple fusion methods. Furthermore, we explore different interaction modes for image selection, finding that a human–computer collaborative approach yields the highest accuracy of 95.4%. This work establishes a framework for AI-assisted breast ultrasound that improves workflow efficiency and diagnostic performance through synergistic human–AI collaboration.

1. Introduction

Breast cancer remains a major public health concern, and accurate diagnosis of breast nodules is paramount for timely and appropriate management, with differentiation between benign lesions and malignant tumors being a critical step. Ultrasound imaging is a widely used, non-invasive, and accessible modality for the initial evaluation of breast nodules [1]. Radiologists and sonographers typically acquire a large series of ultrasound images during a scan session to capture the nodule from multiple angles and demonstrate various features, such as shape, margin, orientation, echo pattern, and posterior acoustic characteristics, as described by the BI-RADS classification system [2].

Historically, the analysis of breast ultrasound (BUS) images for computer-aided diagnosis (CAD) relied heavily on extracting handcrafted features from suspicious regions [3–6]. While these traditional methods provided valuable insights, their performance was often limited by image quality variability and the need for significant domain expertise. The advent of deep learning has revolutionized medical image analysis, enabling models to automatically learn hierarchical and discriminative features directly from raw image pixels, leading to significant improvements in various BUS analysis tasks [7]. Notably, advanced deep learning approaches have shown remarkable success in breast tumor segmentation, addressing challenges such as variable tumor morphology, blurred boundaries, speckle noise, and complex ultrasound patterns through innovative architectures like unpretentious nested U-nets, enhanced selective kernel convolutions, and cascaded convolutional networks [8–10].

Object detection, a fundamental computer vision task involving identifying and localizing objects with bounding boxes [11], has become a crucial application in medical imaging for automating the detection and delineation of abnormalities like lesions [12,13]. Deep learning architectures for object detection are broadly categorized into two-stage (e.g., Faster R-CNN [14]) and one-stage methods (e.g., YOLO, SSD), offering different trade-offs between accuracy and speed.

While subsequent AI-based diagnostic models have shown success in classifying or segmenting nodules on pre-selected images [2,15,16], the upstream process of identifying which images or which specific regions within the numerous scans are most informative for diagnosis has received less attention. The process of selecting the most diagnostically relevant images from potentially hundreds of acquired frames is largely dependent on the operator's experience and subjective judgment. This manual image selection step presents several challenges [17,18], including being time-consuming, inefficient, and prone to inconsistencies due to inter-observer variability.

Furthermore, applying deep learning-based object detection models to BUS, particularly for the purpose of guiding image selection and supporting downstream diagnostic tasks like multimodal analysis,

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presents unique and significant challenges. Ultrasound images often exhibit subtle or low-contrast features and significant variability in lesion appearance. The stringent clinical requirement for high sensitivity necessitates exceptionally robust detection. Beyond these general difficulties, current approaches show specific limitations. Firstly, there is a noticeable gap in effectively combining existing breast ultrasound object detection models with multi-modal analysis frameworks. While multimodal approaches, integrating different data sources like B-mode, elastography, mammography, or clinical records [19–24], are vital for comprehensive diagnosis and risk assessment, they require reliable visual inputs, making the initial image selection step crucial. Recent studies have underscored the power of multimodal fusion, demonstrating enhanced diagnostic outcomes by combining imaging data with clinical and non-imaging information across various contexts, such as breast MRI classification using the MMST-V Transformer architecture [25], liver cancer diagnosis using hybrid fusion of multimodal ultrasound and clinical features [26], and integrating mammography with patient complaints using the MultiBCD model [27]. Modern attention-based architectures, particularly Vision Transformers, have demonstrated exceptional flexibility in seamlessly integrating these disparate data types (such as imaging, scalar measurements, and categorical clinical variables), offering a powerful and systematic solution for robust multimodal analysis in medical diagnostics. The potential of BUS object detection to leverage cine loop features and operate in near real-time could significantly enhance such multimodal pipelines. However, secondly, many existing BUS object detection models are not specifically tailored for processing ultrasound cine loops, lacking specialized feature extraction for temporal dynamics and facing computational challenges for real-time performance necessary for dynamic clinical workflows [28,29]. Thirdly, a lack of robust human–computer collaborative mechanisms limits the practical usability of automated detection tools and inhibits a synergistic workflow. Addressing these challenges is vital for advancing the clinical applicability of object detection in breast imaging for effective image selection and subsequent diagnosis. In response to these limitations and the critical need for automated image selection, we propose a novel deep learning network, termed KIDNet (Key Image Detection Network), specifically designed to automatically detect and highlight key diagnostic regions within breast ultrasound scans of nodules. Leveraging the strengths of the advanced DETR-based framework DEIM-D-FINE [30,31], KIDNet is specifically tailored for breast ultrasound through the incorporation of novel architectural components: a Mixed Aggregation Network (MANet) for enhanced multi-scale feature extraction, Star Block to capture complex spatial dependencies, and Asymmetric Padding Bottleneck (APBottleneck) for optimized receptive field learning in the feature pyramid. By accurately identifying these key regions, KIDNet aims to provide an AI-powered aid to guide physicians in the crucial image selection process, thereby enhancing workflow efficiency and diagnostic consistency, and providing optimal inputs for downstream analyses, including multimodal integration. The primary objective of this study is to develop and evaluate the performance of KIDNet in detecting key diagnostic regions in breast ultrasound images containing nodules, and to demonstrate how this capability facilitates improved downstream diagnostic performance within a multimodal framework. We aim to demonstrate that this automated approach can reliably identify areas that are clinically relevant for diagnosis, laying the groundwork for integrating such a system into the standard breast ultrasound workflow to improve efficiency, reduce subjectivity in image selection, and enhance overall diagnostic accuracy.

2. Results

We structure this section to detail the evaluation outcomes across different components and operational modes. First, we detail the performance of the standalone KIDNet object detection model, including comparisons with baseline methods and ablation studies to validate the contribution of its novel architectural components. Subsequently, we report the diagnostic performance of the multimodal classification model, demonstrating its ability to integrate visual features, KIDNet outputs, and clinical data. Finally, we present the evaluation of different human–computer interaction strategies for image selection on the final patient-level diagnosis accuracy, highlighting the potential benefits of collaborative workflows. All evaluations were conducted on the independent test set.

2.1. Object detection model performance

The performance of KIDNet was evaluated on independent test sets using standardized object detection metrics. The subsequent sections analyze KIDNet's efficacy through comprehensive comparisons with established methodologies. Specifically, we present a quantitative evaluation benchmark comparing KIDNet against both state-of-the-art and baseline object detection models for breast lesion detection. Furthermore, we conducted an extensive ablation study to meticulously examine the individual contributions and synergistic effects of the novel architectural components integrated into KIDNet.

2.1.1. Comparison experiments

The performance of the evaluated object detection models on the internal validation dataset is summarized in Table 1. We assessed the performance using common metrics: Accuracy, Specificity, Sensitivity, and F1-score. Tables 2 and 3 present the performance of these models on the BUSI and BLUI datasets, respectively. It is important to note that the internal dataset is structured such that images are grouped by patient. Following internal guidelines, a patient is classified as malignant if at least one image within their set exceeds the malignancy threshold. In contrast, the external datasets (BUSI and BLUI) are evaluated on a per-image basis.

The detailed confusion matrices for each model, providing insights into true positives, true negatives, false positives, and false negatives, are depicted in Fig. 1. These matrices visually complement the quantitative results, showcasing the classification strengths and weaknesses inherent to each detection architecture. In addition to these metrics, we evaluated the lesion detection performance of the models using Average Precision (AP) at different Intersection over Union (IoU) thresholds: AP@0.5, AP@0.75, and AP@[0.5:0.95]. These metrics, shown in Table 5, provide a more granular assessment of detection accuracy. To better visualize the comparative performance across these AP metrics, a 3D bar chart is presented in Fig. 2. This visualization facilitates a direct comparison of the models, highlighting their performance variations across different IoU thresholds. KIDNet consistently demonstrates superior performance across all datasets and AP metrics, indicating its effectiveness in lesion detection.

Furthermore, to address the feasibility of real-time clinical application, we conducted a comprehensive analysis of the computational performance of KIDNet and other compared models. This analysis includes key metrics such as the number of parameters (Params), Giga Floating Point Operations (GFLOPS), and Frames Per Second (FPS). As presented in Table 4, KIDNet demonstrates a competitive balance between detection accuracy and computational efficiency. With fewer parameters and lower GFLOPS compared to many counterparts, alongside a high FPS, KIDNet shows strong potential for efficient deployment in clinical settings, particularly where rapid processing is crucial.

2.1.2. Ablation study

To investigate the contribution of each novel module in our proposed KIDNet framework, namely the **Mixed Aggregation Network (MA)**, **Star Block (SB)**, and **Asymmetric Padding Bottleneck (APB)**, we conduct an ablation study by selectively removing these components. The results are summarized in Table 6 and visualized in Fig. 3.

From Table 6, we can observe the following:

Table 1
Performance of object detection models on the internal test dataset.

Model	Accuracy	Specificity	Sensitivity	F1-score
YOLOv12	0.874 ± 0.012	0.898 ± 0.017	0.827 ± 0.011	0.816 ± 0.016
SPA	0.874 ± 0.020	0.898 ± 0.020	0.825 ± 0.022	0.815 ± 0.028
RT-DETR v2	0.870 ± 0.008	0.899 ± 0.014	0.813 ± 0.011	0.809 ± 0.009
RetinaNet	0.835 ± 0.020	0.863 ± 0.034	0.778 ± 0.008	0.760 ± 0.021
RT-DETR v1	0.834 ± 0.005	0.866 ± 0.007	0.771 ± 0.029	0.757 ± 0.012
YOLOv8	0.829 ± 0.015	0.857 ± 0.009	0.773 ± 0.033	0.752 ± 0.024
STAC	0.824 ± 0.018	0.849 ± 0.012	0.775 ± 0.050	0.748 ± 0.030
Faster R-CNN	0.822 ± 0.011	0.844 ± 0.014	0.778 ± 0.029	0.746 ± 0.017
KIDNet	**0.902 ± 0.008**	**0.914 ± 0.010**	**0.879 ± 0.019**	**0.858 ± 0.012**

Table 2
Performance of object detection models on the BUSI dataset.

Model	Accuracy	Specificity	Sensitivity	F1-score
RT-DETR v2	0.869 ± 0.020	0.874 ± 0.023	0.859 ± 0.015	0.810 ± 0.026
YOLOv12	0.867 ± 0.027	0.873 ± 0.031	0.854 ± 0.024	0.806 ± 0.037
SPA	0.863 ± 0.009	0.869 ± 0.014	0.851 ± 0.003	0.801 ± 0.009
RetinaNet	0.824 ± 0.016	0.833 ± 0.025	0.806 ± 0.026	0.749 ± 0.019
YOLOv8	0.829 ± 0.019	0.852 ± 0.025	0.781 ± 0.016	0.748 ± 0.024
RT-DETR v1	0.823 ± 0.008	0.844 ± 0.006	0.779 ± 0.034	0.741 ± 0.017
Faster R-CNN	0.802 ± 0.031	0.801 ± 0.037	0.805 ± 0.030	0.726 ± 0.037
STAC	0.805 ± 0.022	0.812 ± 0.031	0.790 ± 0.008	0.725 ± 0.024
KIDNet	**0.888 ± 0.009**	**0.880 ± 0.017**	**0.905 ± 0.008**	**0.840 ± 0.009**

Table 3
Performance of object detection models on the BLUI dataset.

Model	Accuracy	Specificity	Sensitivity	F1-score
YOLOv12	0.886 ± 0.007	0.897 ± 0.034	0.875 ± 0.035	0.879 ± 0.007
RT-DETR v2	0.868 ± 0.007	0.878 ± 0.008	0.856 ± 0.011	0.859 ± 0.007
SPA	0.865 ± 0.025	0.862 ± 0.035	0.869 ± 0.032	0.858 ± 0.025
RetinaNet	0.815 ± 0.023	0.824 ± 0.019	0.804 ± 0.038	0.803 ± 0.027
STAC	0.815 ± 0.030	0.824 ± 0.012	0.804 ± 0.051	0.803 ± 0.035
YOLOv8	0.812 ± 0.014	0.821 ± 0.008	0.801 ± 0.021	0.800 ± 0.016
Faster R-CNN	0.809 ± 0.031	0.824 ± 0.023	0.792 ± 0.041	0.796 ± 0.034
RT-DETR v1	0.806 ± 0.016	0.813 ± 0.037	0.798 ± 0.024	0.795 ± 0.014
KIDNet	**0.895 ± 0.022**	**0.883 ± 0.029**	**0.908 ± 0.016**	**0.891 ± 0.022**

Table 4
Computational performance analysis of different models.

Model Name	Params (M)	GFLOPS	FPS
Faster R-CNN	39.80	180	130
RetinaNet	94.2	375.0	161
YOLOv8	11.2	28.6	311
YOLOv12	9.3	21.4	423
RT-DETR v1	42	136	200
RT-DETR v2	36	100	280
STAC	39.12	160	195
SPA	29.79	120	252
KIDNet	19.85	47.6087	299

Table 5
Lesion detection performance on internal test sets (n = 879 images)

Model	AP@0.5	AP@0.75	AP@[0.5:0.95]
Faster R-CNN	0.721 ± 0.018	0.583 ± 0.024	0.512 ± 0.029
RetinaNet	0.735 ± 0.015	0.601 ± 0.021	0.527 ± 0.027
YOLOv8	0.764 ± 0.014	0.638 ± 0.017	0.562 ± 0.023
YOLOv12	0.778 ± 0.012	0.651 ± 0.016	0.581 ± 0.020
RT-DETR v1	0.792 ± 0.011	0.667 ± 0.014	0.603 ± 0.017
RT-DETR v2	0.811 ± 0.010	0.689 ± 0.013	0.624 ± 0.015
STAC	0.753 ± 0.022	0.612 ± 0.019	0.541 ± 0.026
SPA	0.803 ± 0.011	0.673 ± 0.015	0.609 ± 0.016
KIDNet	**0.853 ± 0.008**	**0.732 ± 0.007**	**0.681 ± 0.006**

- **Impact of Individual Modules:** Adding each module individually (Methods B, C, and D) generally improves performance compared to the baseline (Method A), demonstrating the effectiveness of each component. Specifically, including the **APB** module (Method D) shows a notable improvement in AP@0.5 and AP@[0.5:0.95], suggesting its ability to refine object localization and handle varying object scales.
- **Synergistic Effects:** Combining the modules often yields further improvements. For example, combining the **MA** and **SB** (Method E) results in higher Sensitivity and overall AP scores compared to using either module alone, indicating a positive interaction between these components for better feature representation and spatial understanding.
- **Importance of MA:** Removing the MA (Methods C and G) tends to lead to a decrease in Specificity, suggesting that MA is crucial for effectively fusing multi-scale features and reducing false positives by improving feature discrimination.
- **Importance of SB:** Removing the SB (Methods B and F) generally leads to a reduction in Sensitivity, suggesting that SB is crucial for capturing complex spatial dependencies and enhancing the network's ability to detect true positives, potentially including challenging or subtle nodules.
- **Importance of APB:** When APB is removed (Methods B, C, E), we observe a decrease in AP metrics, reinforcing its role in improving localization accuracy across different Intersection over Union (IoU) thresholds, as highlighted by the improvement in Method D.
- **Overall Best Performance:** The complete configuration with all three modules (Method H) achieves the best performance across all metrics (Specificity, Sensitivity, AP@0.5, and AP@[0.5:0.95]).

Table 6
Ablation study results on the detection task. Performance metrics are reported as mean ± standard deviation.

Method	Module			Spec.	Sens.	AP@0.5	AP@[0.5:0.95]
	MA	SB	APB				
A	–	–	–	0.875 ± 0.012	0.865 ± 0.029	0.800 ± 0.020	0.620 ± 0.025
B	✓	–	–	0.878 ± 0.028	0.875 ± 0.014	0.820 ± 0.018	0.640 ± 0.022
C	–	✓	–	0.864 ± 0.005	0.850 ± 0.045	0.810 ± 0.022	0.630 ± 0.028
D	–	–	✓	0.892 ± 0.031	0.841 ± 0.023	0.830 ± 0.019	0.645 ± 0.023
E	✓	✓	–	0.843 ± 0.017	0.881 ± 0.042	0.845 ± 0.015	0.660 ± 0.020
F	✓	–	✓	0.889 ± 0.045	0.878 ± 0.038	0.835 ± 0.014	0.655 ± 0.018
G	–	✓	✓	0.859 ± 0.005	0.890 ± 0.024	0.840 ± 0.017	0.650 ± 0.024
H	✓	✓	✓	**0.914 ± 0.010**	**0.879 ± 0.019**	**0.853 ± 0.008**	**0.681 ± 0.006**

Table 7
Performance comparison of multimodal classification methods.

Method	Accuracy	Specificity	Sensitivity	F1-Score	AUC
Image Only	82.6 ± 2.1	81.8 ± 4.0	83.5 ± 0.9	81.9 ± 1.8	83.9
Image + Score (MLP)	84.2 ± 1.9	84.0 ± 1.9	84.4 ± 2.4	83.4 ± 2.1	85.1
All modalities (Concat)					
MLP	89.9 ± 1.3	89.7 ± 2.0	90.2 ± 0.5	89.4 ± 1.3	92.1
XGBoost	89.5 ± 0.5	89.2 ± 0.9	89.9 ± 0.0	89.0 ± 0.5	89.3
RandomForest	89.8 ± 1.0	89.4 ± 1.4	90.2 ± 1.9	89.3 ± 1.1	91.4
SVM	89.5 ± 1.4	89.2 ± 1.9	89.9 ± 0.9	89.0 ± 1.4	92.0
LogReg	89.1 ± 1.3	87.8 ± 1.4	90.5 ± 1.9	88.6 ± 1.4	91.1
Proposed model	**93.1 ± 0.4**	**93.8 ± 0.5**	**92.4 ± 0.5**	**92.6 ± 0.5**	**92.7**

Table 8
Diagnostic performance across different operational modes and baseline.

Method/Mode	Accuracy	Specificity	Sensitivity	F1-score
KIDNet baseline	0.902	0.914	0.879	0.858
Automatic selection mode	0.938	0.921	0.957	0.936
Manual selection mode	0.930	0.916	0.945	0.927
Human–Computer collaborative mode	0.954	0.951	0.957	0.951

This strongly suggests that the MA, SB, and APB modules complement each other effectively, leading to a superior overall detection performance for breast nodules.

Fig. 3 provides a visual representation of these trends. It shows that Method H, with all three modules, consistently achieves high relative scores in all metrics on the normalized radar chart, confirming the benefits of the combined approach. The radar chart allows for a clear visual comparison of the performance profiles of different module combinations, highlighting the balanced improvement offered by the full model.

In conclusion, the ablation study clearly demonstrates the positive impact of each individual novel module (MA, SB, and APB) and their synergistic effects in improving the detection performance of our proposed KIDNet framework. The complete configuration (Method H) provides the optimal balance across all evaluated performance metrics, highlighting the effectiveness and necessity of our designed architectural components for accurate breast nodule detection.

2.2. Multimodal classification performance

This section evaluates our Transformer-based multimodal model for breast lesion diagnosis, comparing it against unimodal, bimodal, and simple fusion baselines. We assess performance using Accuracy, AUC, Sensitivity, Specificity, and F1-Score, with full results in Table 7.

Key findings include:

- The image-only baseline achieves 82.6% Accuracy, while adding KIDNet scores (Image + Score) improves to 84.2%.
- Simple concatenation of all modalities (image + score + clinical records) yields 89.1–89.9% Accuracy across classifiers (MLP/XGBoost/RandomForest/SVM/LogReg), confirming clinical data's value.
- Our proposed model outperforms all baselines (93.1% Accuracy, 94.0 AUC), demonstrating superior multimodal fusion.

Fig. 4 combines two complementary visualizations:

- Panel A shows ROC curves, highlighting our model's optimal trade-off between sensitivity/specificity (AUC 94.0).
- Panel B displays normalized metrics across models, confirming consistent superiority in all measures.

2.3. Evaluation of interaction modes

To evaluate the impact of different human–computer interaction modes and the contribution of the multi-modal approach, we compared the diagnostic performance of the system operating in each mode against a baseline model. The baseline utilizes the standalone KIDNet model's output based on a single image (specifically, the image automatically selected by the Automatic Selection Mode logic as the primary input for the CNN), without incorporating additional scores or multimodal fusion.

Fig. 5 illustrates the image selection process in different modes, showing examples of the image automatically selected by the AI-Only logic and the corresponding image after manual or collaborative adjustment. In Case 1, diagnostic accuracy was achieved by selecting the image exhibiting calcification with spiculated irregular morphology over

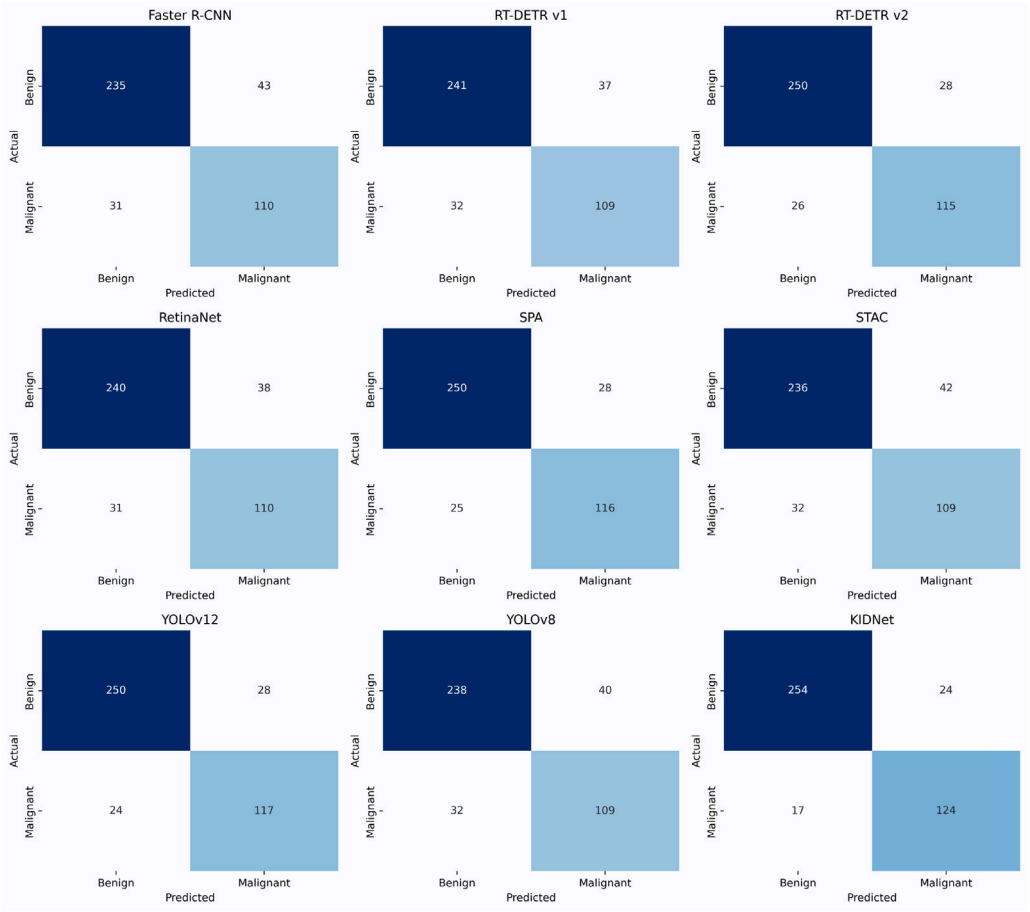


Fig. 1. Confusion matrices for the evaluated object detection models on the internal validation dataset.

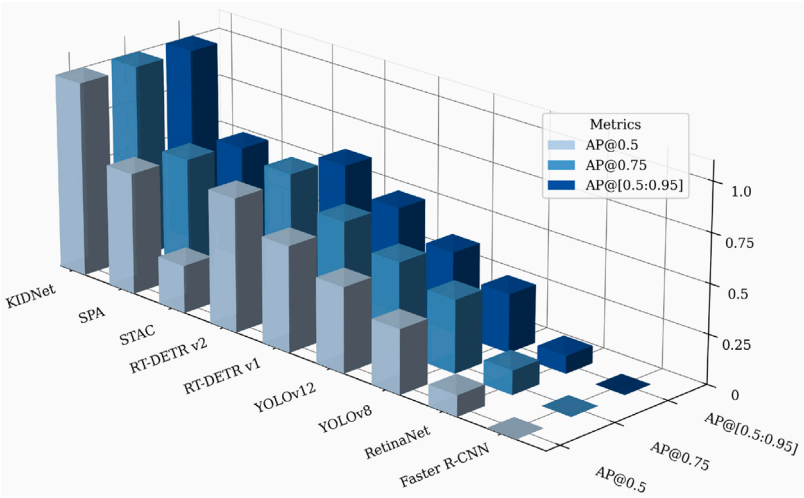


Fig. 2. 3D Bar chart representing the lesion detection performance of different models across various Average Precision (AP) metrics.

two elliptical benign-appearing variants. Case 2 demonstrated improvement through preferential selection of irregular, angular hypo-/anechoic masses rather than their oval counterparts. Cases 3–4 consistently required images maximally displaying irregular lobulated contours and calcifications, rejecting incompletely visualized or less characteristic views. This visual comparison highlights the potential impact of human input on the visual data presented to the system.

Table 8 summarizes the diagnostic performance metrics (e.g., Accuracy, Sensitivity, Specificity, F1-score) for the KIDNet baseline and the system operating in Automatic, Manual, and Human–Computer Collaborative Modes. Fig. 6 provides a visual representation of these results.

Standard deviations are omitted from Table 8. This is due to the experimental design where the Manual Selection Mode was evaluated based on a single experimental session. Additionally, while different physicians participated in the Manual Selection Mode and Human–Computer Collaborative Mode evaluations, their clinical experience and working years were similar to ensure the validity and comparability

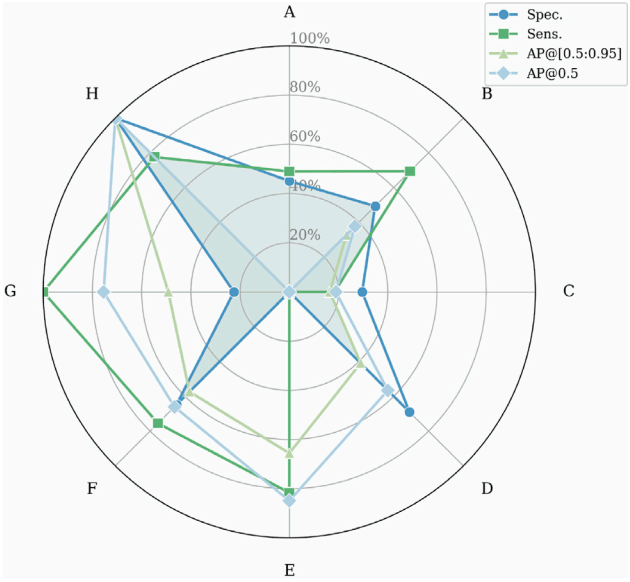


Fig. 3. Radar chart illustrating the impact of different module combinations on performance metrics. The values are Min–Max normalized for visualization across different metrics.

of the results across these modes. A slight discrepancy in performance can be observed between the Manual Selection Mode results presented in this section and those of the Proposed Model. This difference stems from the distinct data sources used for evaluation: the Proposed Model's performance is based on images saved in real-time during clinical diagnosis, while the performance of the Manual Selection Mode evaluated here is based on analysis of archived clinic loops. Furthermore, Fig. 7 provides insight into the model's decision-making process by showing heatmaps that highlight the regions of the image most influential to the prediction. This helps to understand what features the model is focusing on in both the original and potentially adjusted images.

3. Discussion

This study addressed key challenges in breast ultrasound image analysis and diagnosis: the subjectivity and time-consuming nature of manual image selection from cine loops, and the potential to improve diagnostic accuracy by effectively integrating multimodal data and optimizing human–computer interaction. Our work proposed KIDNet, a specialized object detection network for breast lesion detection, and integrated it into a multimodal classification framework that combines visual features, KIDNet-derived scores, and clinical data, designed to function under different human–computer interaction modes.

Our findings demonstrate significant advancements in these areas. First, the developed KIDNet model exhibits promising performance in detecting breast lesions, surpassing several established object detection architectures on both our internal dataset and external public datasets (Tables 1–3 and 5, Fig. 2). The ablation study rigorously confirmed the positive contribution of each proposed novel component — the Mixed Aggregation Network (MA), Star Block (SB), and Asymmetric Padding Bottleneck (APB) — to the overall detection capability (Table 6, Fig. 3). Specifically, the MA enhances feature aggregation from different scales, the SB improves the capture of complex spatial dependencies crucial for irregular lesion shapes, and the APB aids in efficient multi-scale feature learning and precise localization of lesions across varying sizes. The synergistic effect of these modules validates our architectural design choices, demonstrating that tailored architectures can significantly improve performance on challenging medical imaging tasks like BUS lesion detection. This directly addresses the initial limitation discussed in Section 1 by automating the critical first step of efficiently identifying diagnostically relevant regions from potentially extensive cine loops.

Second, our multimodal classification framework effectively leverages complementary information from diverse data sources. As shown in Table 7 and Fig. 4, fusing image features, KIDNet scores, and clinical records substantially improved diagnostic performance compared to unimodal (Image Only) or bimodal (Image + Score) approaches. Importantly, the Transformer-based fusion strategy employed in our proposed model significantly outperformed simpler concatenation approaches, achieving superior metrics across the board, particularly in AUC (Fig. 4). This suggests that the Transformer's attention mechanisms are better suited to model the intricate relationships and interactions between diverse data types (visual features from lesions, quantitative AI scores, and structured clinical data), leading to a more robust and accurate integrated diagnosis. This finding aligns with the increasing recognition of multimodal learning in medical AI and highlights the critical role of sophisticated fusion techniques beyond simple feature stacking.

Third, the evaluation of different human–computer interaction modes provided compelling insights into the optimal integration of AI into the clinical workflow (Table 8, Figs. 5–7). While the Automatic Selection Mode, guided by the KIDNet model's output, showed improved performance compared to a baseline using only a single 'best' image without multimodal fusion, the Human–Computer Collaborative Mode achieved the highest diagnostic accuracy (Accuracy 0.954). This strongly supports the hypothesis that a synergistic approach, combining the AI's efficiency in initial data screening and highlighting potentially significant findings (Fig. 5) with the physician's expert knowledge, contextual understanding, and ability to refine selections based on subtle visual cues or complete clinical context, yields the best outcomes. The Manual Selection Mode also demonstrated value, highlighting the importance of expert image selection, but the collaborative mode surpassed it, suggesting that the AI's ability to provide a ranked list and highlight areas serves as a valuable starting point that can be further optimized by human oversight and decision-making. This finding has significant implications for the design of future AI-assisted diagnostic systems, emphasizing the need for intuitive and effective interfaces that facilitate collaborative decision-making rather than pursuing full automation that bypasses human expertise.

Clinically, the proposed system, particularly operating in the Human–Computer Collaborative Mode, holds significant promise for enhancing breast ultrasound diagnosis. By automating initial lesion detection, it can reduce the time and cognitive load associated with manual image review, allowing radiologists to focus on interpreting key findings. The multimodal fusion approach provides a more comprehensive assessment by integrating diverse patient information. Most importantly, the collaborative mode leverages the strengths of both AI and human expertise, potentially leading to improved diagnostic accuracy and consistency, which could ultimately benefit patient care through earlier and more precise diagnosis.

3.1. Limitations of the study

Despite these encouraging results, several limitations should be acknowledged. Firstly, the evaluation of the Manual Selection Mode was based on a single experimental session, which may not fully capture the inherent variability in manual image selection over time or across a wider range of physicians. This single-session design, coupled with the practical constraints of recruiting multiple equally experienced physicians for repeated trials, also precluded the calculation of standard deviation data for this mode (Table 8), which may limit the robustness of these specific results. While efforts were made to match physician experience for comparability between Manual and Collaborative modes, the use of archived cine loops for evaluating the Manual Mode versus the potential for real-time saved images as input for the Proposed Model evaluation introduces a data source discrepancy that should

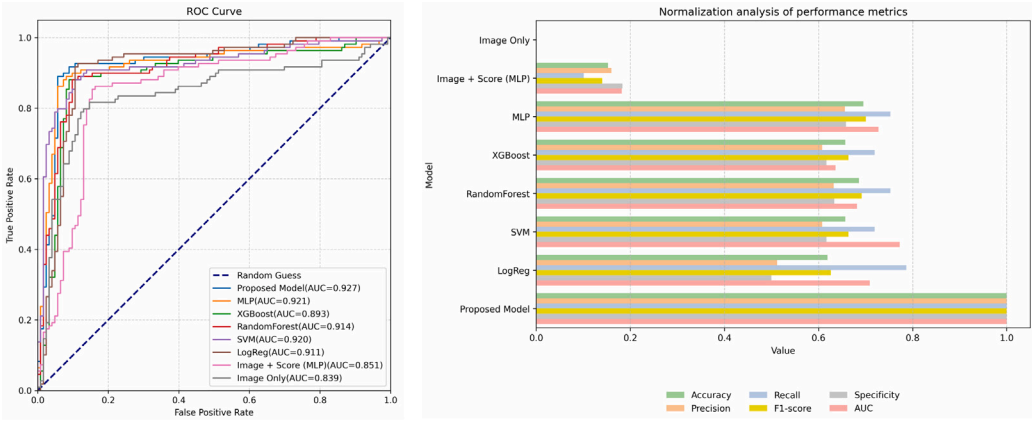


Fig. 4. Multimodal performance analysis: (A) ROC curves with AUC values; (B) Normalized metrics across models.

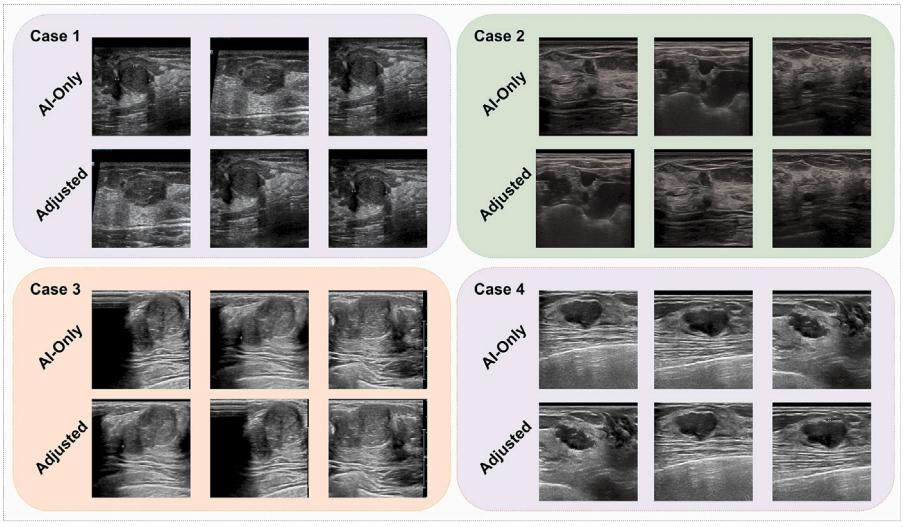


Fig. 5. Examples illustrating image selection in different interaction modes. For each case, the top row shows the image selected by the AI-Only logic (Automatic Selection Mode), while the bottom row shows the corresponding image after manual or collaborative adjustment.

be considered. Future work should aim for prospective evaluation on consistent datasets to address this.

Secondly, while the diagnostic performance of our human–computer interaction models (Collaborative and Manual Modes) was quantitatively assessed through accuracy and efficiency metrics (e.g., time to diagnosis), the evaluation of human factors was not sufficiently comprehensive. Specifically, due to constraints in physician recruitment and available resources, our study currently lacks a systematic quantification of key human–computer interaction metrics, such as physician subjective workload (e.g., using NASA-TLX), decision-making efficiency beyond simple timing, user acceptance, and overall satisfaction/acceptability of the proposed AI-assisted workflow. A formal, large-scale usability study involving clinicians is crucial for assessing the practical workflow integration, user acceptance, and overall user experience of the different interaction modes in a real-world clinical setting. Such a future study would provide valuable, quantifiable insights necessary for optimizing the system's design for clinical translation and fully addressing the benefits of human–computer synergy.

4. Conclusion

In this study, we addressed the challenges in breast ultrasound image analysis and diagnosis by proposing a novel AI-assisted system. We developed KIDNet, a deep learning model specifically designed for accurate and efficient detection of breast lesions within ultrasound images, incorporating novel architectural components like the Mixed Aggregation Network, Star Block, and Asymmetric Padding Bottleneck, which demonstrated significant performance improvements through ablation studies.

We further integrated KIDNet's capabilities into a multimodal classification framework that combines image features, KIDNet-derived malignancy scores, and clinical patient data using a Transformer-based fusion approach. Our results show that this multimodal approach substantially outperforms unimodal and simple concatenation-based baselines in diagnostic accuracy, sensitivity, and specificity, highlighting the power of effectively fusing diverse information sources for breast lesion diagnosis. Crucially, we evaluated the system's performance across different human–computer interaction modes for image selection — Automatic, Manual, and Human–Computer Collaborative — using a

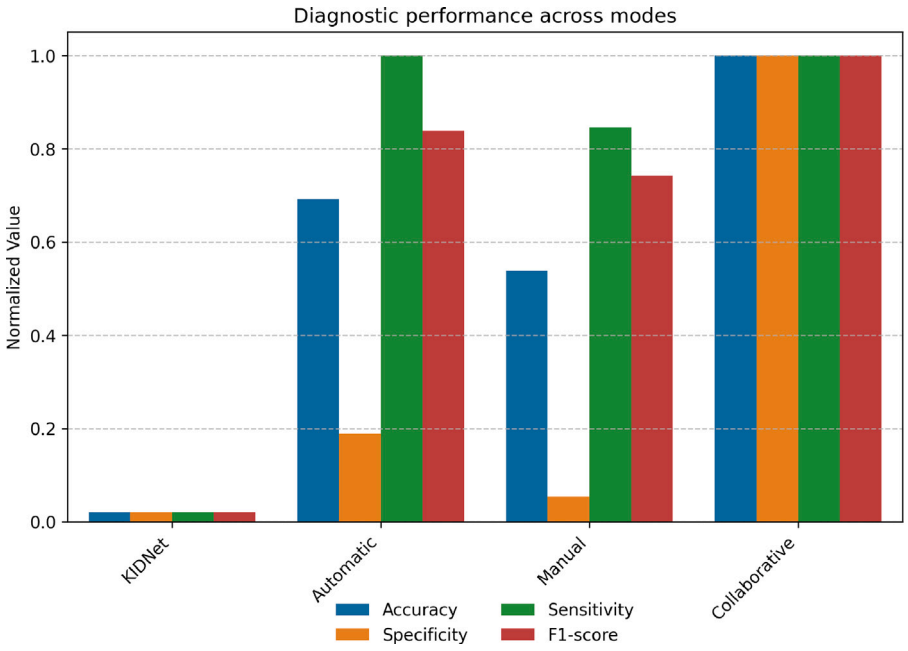


Fig. 6. Normalized performance metrics (Accuracy, Specificity, Sensitivity, F1-score) for the KIDNet baseline and the system operating in Automatic, Manual, and Human–Computer Collaborative Modes. This bar chart visualizes the data presented in Table 8.

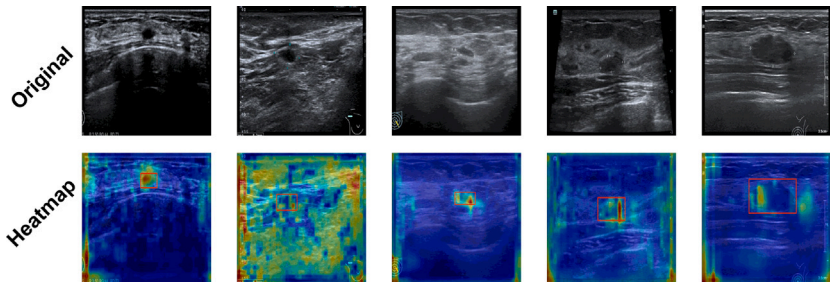


Fig. 7. Examples illustrating model focus via heatmaps. The top row shows original ultrasound images, and the bottom row displays corresponding heatmaps indicating the regions the model focused on for prediction.

dedicated dataset reflecting clinical workflow. The Human–Computer Collaborative Mode achieved the highest diagnostic performance metrics, demonstrating that the synergistic combination of AI's ability to identify and rank potentially important images with the physician's expert review and adjustment leads to superior diagnostic outcomes.

This work provides a robust framework for an AI-assisted breast ultrasound system that can enhance diagnostic workflow efficiency by automating key region identification, improve diagnostic accuracy through multimodal data integration, and promote effective human–AI collaboration. Our findings emphasize that the future of medical AI lies not in complete automation, but in creating intelligent tools that augment human expertise, leading to more reliable and consistent clinical decisions.

5. Main tables

Standard deviations are omitted from Table 8. This is due to the experimental design where the Manual Selection Mode was evaluated based on a single experimental session. Additionally, while different physicians participated in the Manual Selection Mode and Human–Computer Collaborative Mode evaluations, their clinical experience and working years were similar to ensure the validity and comparability of the results across these modes.

Table 9
Baseline characteristics of study participants.

Patients	Malignant mean	Benign mean	P-Value
Age (years)	38.9	47.2	0.001
BMI (kg/m^2)	23.41	22.19	0.343
CA125 (U/mL)	29.10	23.31	0.696
CA153 (U/mL)	25.28	17.41	0.383
CA199 (U/mL)	67.99	34.71	0.542
CEA (ng/mL)	11.62	5.46	0.311
AFP (ng/mL)	6.44	2.55	0.631

Table 10
Dataset composition and class distribution.

Set	Cases (n)	Images (n)	B/M (Cases)	B/M (Images)
Training	1409	4120	935:474	2720:1,400
Validation	273	851	181:92	570:281
Internal test	419	1344	278:141	885:459
External	**N/A**	**879**	**N/A**	**546:333**
BUSI	–	647	–	437:210
BLUI	–	232	–	109:123
Total	2101	7194	1399:702	4175:2,473

Note: B=Benign, M=Malignant. Internal data were split at patient level. External sets maintained original image-level annotations.

Table 11
Training hyperparameters for KIDNet and multimodal model.

Hyperparameter	KIDNet	Multimodal model
Epochs	132	50
Optimizer	AdamW	AdamW
Initial learning rate	4×10^{-4}	1×10^{-4}
Weight decay	0.01	0.01
Batch size	16	32
Loss function	Cross-Entropy loss	Cross-Entropy loss
Learning rate Schedule	MultiStepLR	–

CRediT authorship contribution statement

Yue Zhai: Formal analysis, Data curation, Conceptualization. **Dianhuan Tan:** Methodology. **Desheng Sun:** Resources, Funding acquisition, Conceptualization. **Tingting Zheng:** Funding acquisition.

Ethics approval

The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki. The study was approved by the Medical Ethics Committee of Peking University Shenzhen Hospital (Approval Number: Peking University Shenzhen Hospital Ethics Committee (Research) [2022] No. 136), and the requirement of individual consent for this retrospective analysis was waived.

Declaration of generative AI use

During the preparation of this work, the author(s) used DeepSeek-V3 in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Methods

A.1. Key resources table

We have used computational resources, software libraries, and publicly available datasets in this study. A detailed list of these resources, including hardware specifications, software versions, and datasets, is provided in the Key Resources Table, which will be uploaded separately.

A.2. Experimental model and study participant details

Ethical approval for this study was obtained from the Institutional Review Ethics Committee of Peking University Shenzhen Hospital, China.

This study constructed a multimodal dataset derived from **2101 patients** who underwent breast ultrasound examinations at our institution between **May 2022 and December 2023**. The dataset comprises clinical records, a total of **419 cine loops**, and **6315 static ultrasound images**. These images were acquired using a variety of ultrasound diagnostic systems, including but not limited to: **GE E9, Philips Q5, Hitachi Ascendus, Toshiba Aplio 400, Mindray DC-8, Mindray Resona 70B**, among other models.

Study cohort selection criteria. To ensure data quality and study validity, subjects were required to meet the following inclusion criteria and excluded based on the listed conditions:

Inclusion criteria:

1. Patients who underwent breast ultrasound examination.
2. Availability of clear breast ultrasound images, demonstrating distinct lesions with complete contours.
3. Pathological confirmation of a breast tumor.
4. Availability of complete clinical information, including age, BMI, and results for tumor markers.

Exclusion criteria:

1. Poor ultrasound image quality or incomplete lesion visibility precluding accurate imaging analysis.
2. Absence of a definitive pathological diagnosis for the breast lesion.
3. History of prior breast surgery or other treatments that could influence ultrasound appearance or tumor marker levels.

Fig. 8 illustrates the process of patient enrollment and selection following the application of the inclusion and exclusion criteria. The baseline characteristics of the study participants included in the analysis are summarized in Table 9.

Experimental models. The following object detection models were used in this study:

- **YOLOv12** [32]: An attention-centric real-time object detector.
- **RetinaNet** [33]: A single-stage object detector using a Focal Loss to address class imbalance.
- **Faster R-CNN** [14]: A classical supervised object detection framework with ResNet as a backbone, which employs a region proposal network to generate regions of interest (ROIs).
- **YOLOv8** [34]: A state-of-the-art object detector.
- **RT-DETR v1** [35]: An efficient hybrid encoder.
- **RT-DETR v2** [36]: An improved baseline with bag-of-freebies for Real-Time Detection Transformer.
- **STAC** [37]: STAC was selected for comparison as the classical semi-supervised-learning-based detection and classification, in which the self-training and augmentation-driven consistency regularization were adopted to generate pseudo-ROIs on image-level data.

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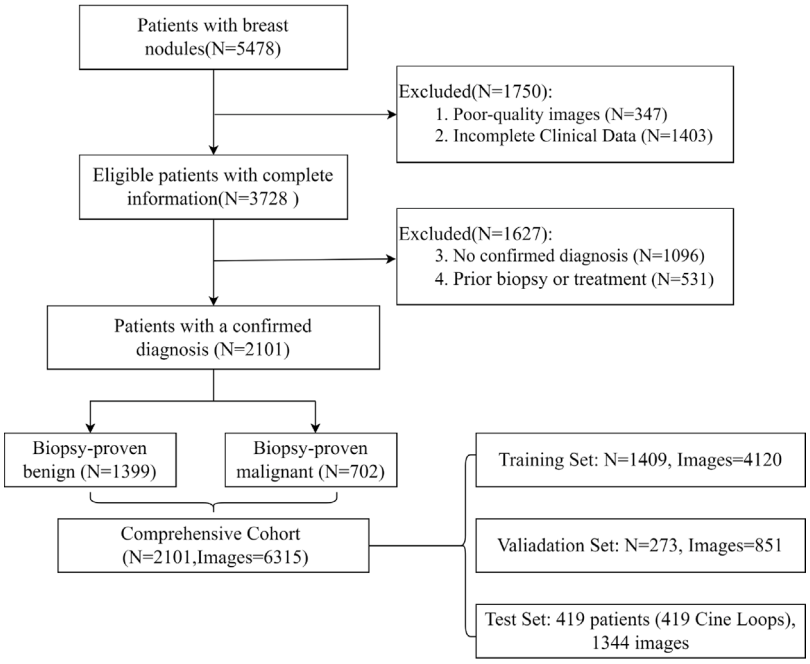


Fig. 8. Flowchart of patient enrollment, selection, and dataset partitioning.

Public datasets for external validation. To further validate the generalizability of our proposed method, we also evaluated it on two publicly available breast ultrasound datasets: BUSI and BLUI.

BUSI dataset The Breast Ultrasound Images (BUSI) dataset was collected from 600 female patients aged 25 to 75 years at Baheye Hospital in 2018 [38]. This dataset comprises 780 ultrasound images with an average size of 500 × 500 pixels, categorized as normal, benign, or malignant. We utilized the 647 images labeled as benign (437 images) or malignant (210 images) for testing. Region-of-interest (ROI) level labels were manually annotated by a sonologist for these images.

BLUI dataset The Breast Lesion Ultrasound Images (BLUI) dataset was acquired from Shahid Beheshti University of Medical Sciences, Tehran, Iran [39]. It contains 232 breast ultrasound images, consisting of 109 benign and 123 malignant cases. All images were annotated by a board-approved radiologist with 20 years of experience, based on histopathologic study results (fine needle aspiration, core needle, or open biopsies).

A.3. Method details

Data preprocessing and annotation. The preparation of data for both model training phases relies on the core dataset derived from the 2101 patients. For ultrasound images, the Region of Interest (ROI) containing the breast lesion was extracted based on annotations. These lesion location annotations were meticulously performed and reviewed by two experienced ultrasound physicians, each with over three years of clinical experience. Pathology classification labels (benign/malignant) were extracted directly from the corresponding pathology reports.

Proposed KIDNet architecture. This section presents the proposed KIDNet architecture for Breast Ultrasound lesion detection, illustrated in Fig. 9. KIDNet is designed to be efficient and effective, building upon established frameworks like DEIM and D-FINE with tailored architectural components for the specific task of BUS detection.

As depicted in Fig. 9, the architecture follows a standard object detection paradigm comprising a backbone network for hierarchical feature extraction, followed by a Feature Pyramid Network (FPN) and a Path Aggregation Network (PAN) for multi-scale feature fusion, and an implicit prediction head.

The backbone, termed **MANStar HGNetv2**, is a modification of the standard HGNetv2 structure. It processes the input via a **Stem** and sequential Stages (Stage 1–4). As depicted in Fig. 9 detailing the **HG-Stage** and **MANStar-HG** components, the core modification involves replacing standard HG blocks with the proposed **MANStar Block**. This **MANStar Block** integrates MANet and Star modules for efficient and potentially enhanced feature representation tailored for medical imaging. These **MANStar Block**s are typically organized within **MANStar-HG** units, which are key building blocks of the modified stages (**HG Stages**).

The multi-scale features extracted by the backbone are then fed into the FPN and PAN. These components are responsible for aggregating semantic information from deeper layers and spatial information from shallower layers across different scales to enrich the feature maps for detection. A key architectural modification in our proposed KIDNet, visualized in the FPN/PAN details and the **C2f** structure in Fig. 9, is the replacement of traditional fusion blocks (like CSPLayer or RepNCSPELAN) with the **C2f Block** that incorporates **APBottleneck** (Asymmetric Padding Bottleneck). This design choice is motivated by the need for a more lightweight yet effective cross-stage feature fusion, offering improved efficiency and potentially better performance compared to its predecessors, suitable for real-time applications.

Furthermore, KIDNet incorporates the **DEIM training framework**. Addressing the challenge of sparse supervision and limited positive samples typical in BUS datasets, DEIM employs a **Dense O2O matching strategy**. This strategy enhances the number of positive targets per image by leveraging data augmentation, thereby enriching the training signal and better matching the characteristic of positive sample scarcity in BUS data.

The prediction head incorporates key ideas from **D-FINE**, including **Feature Distribution Refinement (FDR)** and **Global Optimal Localization Self-Distillation (GO-LSD)**. **FDR** models boundary predictions as probability distributions to capture positional uncertainty, which is particularly beneficial for the often ambiguous BUS lesion boundaries. **GO-LSD** acts as a self-distillation mechanism during training, transferring refined localization knowledge from deeper layers to shallower ones, improving early layer learning and overall accuracy.

In summary, KIDNet integrates a **MANStar HGNetv2** backbone, an optimized FPN/PAN neck utilizing **C2f Block** with **APBottleneck**,

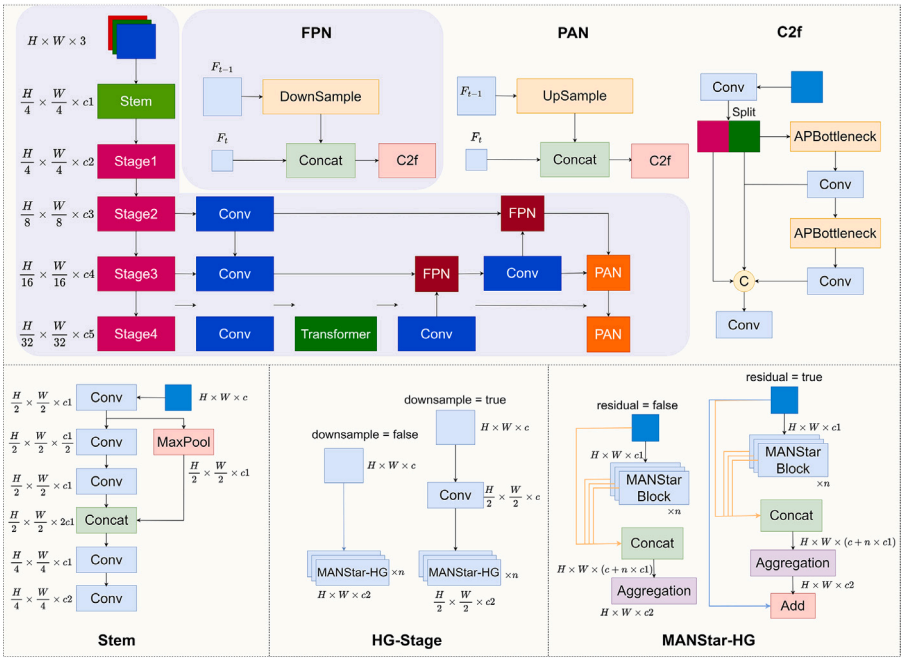


Fig. 9. Overall architecture of the proposed KIDNet. The network comprises a **MANStar HGNetv2** backbone for hierarchical feature extraction, and a neck composed of FPN and PAN utilizing **C2f Block** with **APBottleneck** for multi-scale feature fusion. Key DEIM and D-FINE concepts, including dense matching, probability distribution refinement, and self-distillation, are incorporated into the training framework and prediction head.

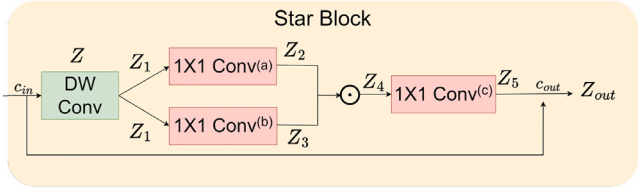


Fig. 10. Structure of the Star Block. The 1×1 Convs are Pointwise Convolutions, where the superscripts $(a), (b), (c)$ denote three distinct sets of learned parameters.

and the powerful DEIM and D-FINE training and prediction strategies, resulting in an efficient and robust architecture specifically designed for BUS lesion detection.

Star block. As detailed in the previous subsection, the MANet incorporates a "Cascaded Star Block Path" where features are processed through a sequence of n cascaded sub-modules, specifically **Star Block**s. The Star Block is designed to capture complex spatial dependencies efficiently and serves as a key component for enriching the feature representations. Its structure is illustrated in Fig. 10.

Given an input feature map $Z \in \mathbb{R}^{H \times W \times c}$ to a Star Block (where c is the number of channels), the block first applies a depthwise convolution (DWConv):

$$Z_1 = \text{DWConv}_{k \times k}(Z) \tag{1}$$

The output Z_1 is then processed by two parallel branches, each consisting of a 1×1 convolution. **Note that the superscripts (a) and (b) indicate that these branches utilize distinct and independent convolution parameters, typically used for channel expansion.**

$$Z_2 = \text{Conv}^{(a)}_{1\times1}(Z_1) \tag{2}$$

$$Z_3 = \text{Conv}^{(b)}_{1\times1}(Z_1) \tag{3}$$

The outputs of these two parallel 1×1 convolutions, Z_2 and Z_3, are then combined using element-wise multiplication ($\odot$), **which serves as a non-linear feature gating mechanism** (similar to a Gated Linear Unit, GLU):

$$Z_4 = Z_2 \odot Z_3 \tag{4}$$

The result Z_4 then passes through a final 1×1 convolution, $\text{Conv}^{(c)}$, which projects the features back to the required channel dimension:

$$Z_5 = \text{Conv}^{(c)}_{1\times1}(Z_4) \tag{5}$$

Finally, a residual connection is added, summing the output Z_5 with the original input Z:

$$Z_{\text{out}} = Z_5 + Z \tag{6}$$

This residual connection helps mitigate the vanishing gradient problem and facilitates easier training of deep networks. By combining depthwise convolution, feature expansion using independent 1×1 convolutions, the non-linear element-wise multiplication for feature interaction (gating), and a residual connection, the Star Block effectively captures intricate spatial correlations and dependencies efficiently.

Mixed Aggregation Network (MANet). To enhance the feature extraction capabilities of the backbone network, we introduce the Mixed Aggregation Network (MANet) block, as illustrated and annotated in Fig. 11. MANet serves as the core building block within the stages of our modified HGNetv2 backbone (MANStar HGNetv2), replacing standard modules. The basic convolutional unit used throughout is the $k \times k$ Conv structure (Conv + BN + SiLU), defined on the right side of the figure.

The design of MANet is based on aggregating features processed through multiple parallel and sequential paths. Given an input feature map $X_{\text{in}} \in \mathbb{R}^{H \times W \times c_{\text{in}}}$ (where H and W are height and width), the module first applies a 1×1 convolution to project the input to an intermediate representation $\mathbf{Y}$ with expanded channel dimension $2c$:

$$\mathbf{Y} = \text{Conv}_{1\times1}(X_{\text{in}}), \quad \mathbf{Y} \in \mathbb{R}^{H \times W \times 2c} \tag{7}$$

As shown in Fig. 11, the tensor $\mathbf{Y}$ is then processed through four main parallel paths ($\mathbf{Y_{P1}}, \mathbf{Y_2}, \mathbf{Y_1}, \mathbf{Y_3}$), all derived from $\mathbf{Y}$:

- **Path P1 (Auxiliary 1×1 Path, $\mathbf{Y_{P1}}$):** $\mathbf{Y}$ passes through a 1×1 convolution (indicated by the dashed connection above Y) to

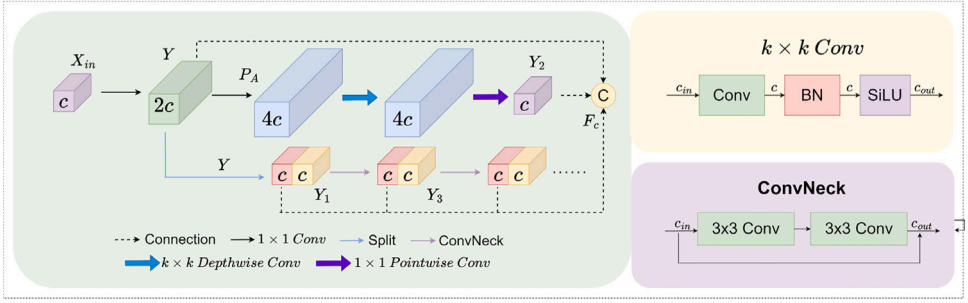


Fig. 11. Illustration of the Mixed Aggregation Network (MANet) block structure within the MANStar-HG/HG-Stage. Key feature maps are labeled $X_{\text{in}}, Y, Y_1, Y_2, Y_3, Y_4, Y_c$. The legend clarifies the arrow types and symbols.

reduce its channel dimension from $2c$ to c. This output, $\mathbf{Y}_{\text{P1}} \in \mathbb{R}^{H\times W\times c}$, is collected for final concatenation.

$$\mathbf{Y}_{\text{P1}} = \text{Conv}_{1\times1}(\mathbf{Y}) \tag{8}$$

- **Path P2 (Depthwise Separable Convolution Path, $\mathbf{Y_2}$):** $\mathbf{Y}$ is processed by a three-layer Depthwise Separable Block (via P_A blocks), which consists of a pointwise convolution, a depthwise convolution, and a final pointwise convolution, producing output $\mathbf{Y_2} \in \mathbb{R}^{H\times W\times c}$.

$$\mathbf{Y_2} = \text{Conv}_{1\times1}(\text{DWConv}_{k\times k}(\text{Conv}_{1\times1}(\mathbf{Y}))) \tag{9}$$

- **Path P3 & P4 (Split Paths, $\mathbf{Y_1}$ and $\mathbf{Y_3}$):** The tensor $\mathbf{Y}$ is split evenly along the channel dimension into two halves, $\mathbf{Y_1}$ and $\mathbf{Y_3}$ (labeled Y_1, Y_3 in the figure, each $\in \mathbb{R}^{H\times W\times c}$).

$$(\mathbf{Y_1}, \mathbf{Y_3}) = \text{Split}(\mathbf{Y}) \tag{10}$$

  - **Path P3 Short-cut ($\mathbf{Y_1}$):** The first half, $\mathbf{Y_1}$, serves as a direct short-cut connection to the aggregation point.
  - **Path P4 Cascaded Star Block ($\mathbf{Y_3}$ and $\mathbf{Y_{4,i}}$):** The second half, $\mathbf{Y_3}$, is collected for aggregation and also serves as the input to a cascade of n Star Block modules. (The Star Block is based on the ConvNeck structure defined in the figure, and its details are presented in the next subsection). The output of the ith Star Block is denoted $Y_{4,i}$ $(1 \le i \le n)$, all $\in \mathbb{R}^{H\times W\times c}$.

$$Y_{4,1}, \ldots, Y_{4,n} = \text{StarBlocks}^{(n)}(\mathbf{Y_3}) \tag{11}$$

The outputs from all $4 + n$ features–$\mathbf{Y}_{\text{P1}}$, $\mathbf{Y_2}$, $\mathbf{Y_1}$ (Split 1), $\mathbf{Y_3}$ (Split 2), and the n sequence outputs $Y_{4,1}$ to $Y_{4,n}$–are then concatenated along the channel dimension (Y_c).

$$\mathbf{F}_{\text{concat}} = \text{Concat}(\mathbf{Y}_{\text{P1}}, \mathbf{Y_2}, \mathbf{Y_1}, \mathbf{Y_3}, Y_{4,1}, \ldots, Y_{4,n}), \quad F_{\text{concat}} \in \mathbb{R}^{H\times W\times(\mathbf{4+n})\mathbf{c}} \tag{12}$$

Finally, a 1×1 convolution projects the concatenated features back to the desired output channel size c_{out}, yielding the final output of the MANet block:

$$Y_{\text{out}} = \text{Conv}_{1\times1}(F_{\text{concat}}), \quad Y_{\text{out}} \in \mathbb{R}^{H\times W\times c_{\text{out}}} \tag{13}$$

This mixed aggregation strategy, leveraging diverse computational paths (including 1×1 projections, depthwise separability, direct pathways, and cascaded ConvNeck structures), ensures a rich and diverse feature representation, enhancing gradient flow and semantic depth.

Asymmetric Padding Bottleneck (APBottleneck). The Asymmetric Padding Bottleneck is a novel module designed to enhance the capture of diverse spatial features by employing asymmetric padding before standard convolutions. This approach allows the network to expand its receptive field in different directions across parallel branches. It is utilized as a core building block (e.g., within C2f structures) in the architecture. Its structure is illustrated in Fig. 12.

Given an input feature map $X \in \mathbb{R}^{H_1\times W_1\times c_1}$ (where c_1 is the input channel dimension), the APBottleneck performs the following operations:

1. **Asymmetric Padding**: The input feature map X is padded asymmetrically in four distinct ways according to configurations like $(1, 0, 0, 3)$, $(0, 3, 0, 1)$, $(0, 1, 3, 0)$, and $(3, 0, 1, 0)$.
2. **Convolutional Layers**: The padded feature maps are then passed through four parallel convolutional layers. Each layer consists of a convolutional operation with a kernel size of 1×3 or 3×1 with a stride of 1, followed by batch normalization.
3. **Concatenation**: The outputs of the four batch-normalized convolutional layers are then concatenated along the channel dimension.
4. **Output Transformation**: Finally, a linear transformation is applied to reduce the channel dimension, followed by batch normalization.

The APBottleneck's strategic use of asymmetric padding and parallel directional convolutions enables it to capture fine-grained spatial information efficiently and is particularly suited for capturing features of varying scales in medical images.

Integration with multi-modal analysis. Our proposed system leverages a multimodal approach to enhance the accuracy of breast lesion diagnosis in ultrasound images. As illustrated in Fig. 13, the model integrates information from three distinct modalities: image data, deep learning-derived scores from KIDNet, and clinical records.

The processing flow is as follows:

1. **Image Data Processing**: Three ultrasound images, each with dimensions 224×224x3, serve as input to a CNN backbone. This CNN extracts relevant visual features, generating a feature map of size 128×56x56.
2. **KIDNet Score Integration**: The same three ultrasound images are also processed by the KIDNet lesion detection and classification model. KIDNet outputs a malignancy confidence score (malignant class confidence minus benign class confidence) for each image.
3. **Clinical Record Incorporation**: Clinical records, including features such as patient age, tumor size, and serum marker information, are incorporated. These features are first normalized using Min–Max scaling.
4. **Feature Fusion**: The CNN Backbone output is divided into image patches and flattened for Patch Embedding. MLP Layers process the KIDNet output and normalized Clinical Records to a Projected Feats vector. A learnable Cls Token is used. A Transformer encoder merges features from all three pathways (Patch Embedding, Projected Feats, Cls Token).
5. **Classification**: The fused feature representation from the Transformer is then passed to a classifier, which outputs the final prediction of lesion malignancy (benign or malignant).

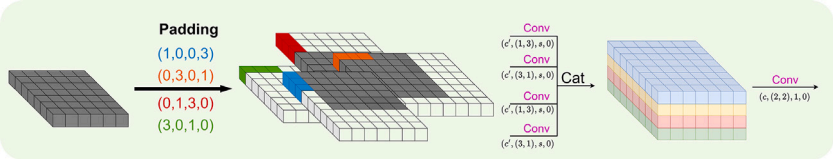


Fig. 12. Structure of the asymmetric padding bottleneck.

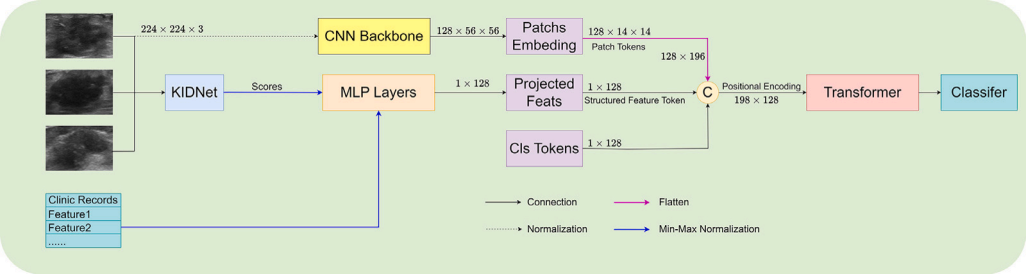


Fig. 13. Structure of the multimodal analysis model.

The combination of these modalities allows the model to leverage complementary information from visual features, a pre-trained deep learning assessment, and patient-specific contextual information.

Human–computer interaction aspects. The system provides three operational modes for clinicians to interact with the model, primarily influencing the selection of input images for the multimodal analysis:

1. **Manual Selection Mode**: The clinician manually selects three ultrasound images. The clinician chooses the most representative image from these three for input into the CNN backbone. The KIDNet malignancy confidence scores for all three selected images are used as input to the MLP Layers.
2. **Automatic Selection Mode**: The system automatically processes a cine loop, records KIDNet confidence scores per frame, and selects the three images with the highest confidence difference scores, avoiding redundancy within a 30-frame window. The image with the highest malignancy confidence is used for the CNN backbone, and the confidence scores of the three images are used as input to the MLP Layers.
3. **Human–Computer Collaborative Mode**: The system initially selects three images automatically. The clinician reviews these images and can adjust their order or replace them with others from the cine loop. The final set of three images (potentially adjusted by the clinician) determines the input for the CNN backbone (highest confidence image) and the KIDNet scores for the MLP Layers.

These modes allow tailoring the system's operation to specific needs, balancing expert judgment, automation, and collaborative refinement.

Dataset usage for model training. Model training was conducted in two sequential stages, each utilizing the prepared dataset with specific input configurations:

1. **KIDNet Training:** The initial stage involved training a standalone KIDNet model. Extracted lesion ROIs were resized to 640 × 640 pixels. Training utilized annotated lesion location as localization ground truth and pathology labels for classification.
2. **Multimodal Model Training:** Following KIDNet training, the multimodal analysis model was trained. Inputs were extracted lesion ROIs resized to 224 × 224 pixels (for the CNN Backbone) and numerical features derived from clinical records and KIDNet scores (for the structured input branch). All structured features underwent Min–Max scaling globally across the 2101-patient cohort.

Dataset usage for human–computer collaboration validation. Evaluation of human–computer collaborative diagnosis strategies was conducted exclusively on the dedicated test set (comprising 419 patients). The difference lay in the method of selecting the specific images used as input for the multimodal model's visual branch (and consequently, influencing the KIDNet scores used as structured features):

1. **Manual Image Selection:** A single ultrasound physician reviewed cine loops for each lesion in the test set (allowed review twice) and manually selected the three most representative or informative images.
2. **Automatic Image Selection:** The multimodal model itself automatically selected the three most relevant images from the available images for each lesion based on an internal criterion.
3. **Hybrid Image Selection:** The model initially selected three images automatically. The human physician then reviewed these and had the option to adjust their order or replace them, determining the final set of three images used as visual inputs.

For these validation scenarios, only the selected images (specifically the top-ranked three) were fed into the CNN Backbone of the multimodal model. The KIDNet scores corresponding to these selected images were utilized as part of the structured input features.

Model implementation details. The multimodal model architecture is described above. For the image processing branch, a pre-trained ResNet18 model, initialized with ImageNet weights, was used. The global average pooling and fully connected layers were removed, and its output feature maps were further processed. ResNet18 weights were frozen during training.

Structured data underwent Multi-Layer Perceptron (MLP) processing for embedding. Image feature maps were divided into patches and linearly embedded. A learnable classification token was prepended to the sequence of embedded image patches and projected structured features. This combined sequence served as input to a Transformer encoder consisting of 4 layers, with 8 attention heads and a hidden dimension of 128. Final classification was by a linear layer on the classification token's output embedding.

A.4. Quantification and statistical analysis

All experiments were conducted on a computing platform equipped with an NVIDIA GeForce RTX 3090 GPU and a Intel Xeon(R) Platinum 8362 CPU. The models were implemented and trained using Python 3.12 with the PyTorch deep learning framework (version 2.1.2). Key libraries utilized include torchvision, NumPy, Pandas, and Scikit-learn.

Dataset splitting. For all training and evaluation procedures, the 2101-patient cohort was divided into training, validation, and test sets at the **patient level** to prevent data leakage. To specifically support subsequent simulations of real-time human–computer collaborative diagnosis, patients with clearly defined 'cline loops' — distinct imaging features crucial for diagnostic assessment and workflow integration — were prioritized for inclusion in the test set. The split was performed using stratified random sampling based on lesion diagnosis (benign/malignant). To ensure model generalizability, two external validation sets were incorporated: BUSI (647 test images) and BLUI (232 test images). The final dataset split statistics are summarized in Table 10.

Training configuration. The overall model training process was conducted in two distinct phases: training of the standalone KIDNet model and subsequent training of the multimodal analysis model. Each phase had specific training configurations, as comprehensively summarized in Table 11. The training objective for both models was to minimize the Cross-Entropy Loss for their respective classification tasks. For both training processes, model checkpoints were saved based on achieving the lowest validation loss during the training process for each respective model.

Evaluation metrics. The performance of the developed models was rigorously evaluated on an independent test set. Appropriate evaluation metrics were employed based on the task performed by the models.

Common binary classification metrics For stages involving binary classification (benign/malignant prediction), standard metrics derived from the Confusion Matrix were used, based on True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN):

- **Accuracy**: $\frac{TP+TN}{TP+TN+FP+FN}$
- **Sensitivity/True Positive Rate (TPR)**: $\frac{TP}{TP+FN}$ (if $TP+FN>0$)
- **Specificity**: $\frac{TN}{TN+FP}$ (if $TN+FP>0$)
- **F1-score**: $2\times\frac{\text{Sensitivity}\times\text{Specificity}}{\text{Sensitivity}+\text{Specificity}}$ (if Sensitivity + Specificity > 0)
- **Confusion Matrix**: Table summarizing TP, TN, FP, FN.
- **ROC Curve and AUC**: ROC curve plots TPR vs. FPR ($FPR=\frac{FP}{FP+TN}$), AUC is the area under the curve.

Object detection metrics For the object detection model, metrics evaluating localization and classification accuracy of detected bounding boxes are used, based on Intersection over Union (IoU): $IoU=\frac{\text{Area of Intersection}}{\text{Area of Union}}$.

- **Average Precision (AP)**: Area under the Precision-Recall curve for the "lesion" class.
- **AP@0.5**: AP calculated using an IoU threshold of 0.5.
- **AP@0.5–0.95**: Average AP across IoU thresholds from 0.5 to 0.95 with steps of 0.05.

Data availability

• The data used in this study contain sensitive patient information and are therefore not publicly available to protect patient privacy.
• The code for this study is openly available at https://github.com/ychy7001/KIDNet.
• Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Desheng Sun.

Materials availability

This study did not generate new materials.

References

[1] Santeramo R, Damiani C, Wei J, Montana G, Brentnall AR. Are better AI algorithms for breast cancer detection also better at predicting risk? A paired case-control study. Breast Cancer Res : BCR 2024;26(1):25. http://dx.doi.org/10.1186/s13058-024-01775-z.

[2] Hameed Z, Garcia-Zapirain B, Aguirre JJ, Isaza-Ruget MA. Multiclass classification of breast cancer histopathology images using multilevel features of deep convolutional neural network. Sci Rep 2022;12(1):15600. http://dx.doi.org/10.1038/s41598-022-19278-2.

[3] Yu F-H, Miao S-M, Li C-Y, Hang J, Deng J, Ye X-H, Liu Y. Pretreatment ultrasound-based deep learning radiomics model for the early prediction of pathologic response to neoadjuvant chemotherapy in breast cancer. Eur Radiol 2023;33(8):5634–44. http://dx.doi.org/10.1007/s00330-023-09555-7.

[4] Ferro A, Bottosso M, Dieci MV, Scagliori E, Miglietta F, Aldegheri V, Bonanno L, Caumo F, Guarneri V, Griguolo G, Pasello G. Clinical applications of radiomics and deep learning in breast and lung cancer: A narrative literature review on current evidence and future perspectives. Crit Rev Oncol Hematol 2024;203:104479. http://dx.doi.org/10.1016/j.critrevonc.2024.104479.

[5] Qi Y-J, Su G-H, You C, Zhang X, Xiao Y, Jiang Y-Z, Shao Z-M. Radiomics in breast cancer: Current advances and future directions. Cell Rep Med 2024;5(9):101719. http://dx.doi.org/10.1016/j.xcrm.2024.101719.

[6] Lee SH, Park H, Ko ES. Radiomics in Breast Imaging from Techniques to Clinical Applications: A Review. Korean J Radiol 2020;21(7):779–92. http://dx.doi.org/10.3348/kjr.2019.0855.

[7] Eghtedari M, Chong A, Rakow-Penner R, Ojeda-Fournier H. Current Status and Future of BI-RADS in Multimodality Imaging, From the AJR Special Series on Radiology Reporting and Data Systems. AJR Am J Roentgenol 2021;216(4):860–73. http://dx.doi.org/10.2214/AJR.20.24894.

[8] Chen G, Li L, Zhang J, Dai Y. Rethinking the unpretentious U-net for medical ultrasound image segmentation. Pattern Recognit 2023;142:109728. http://dx.doi.org/10.1016/j.patcog.2023.109728, URL https://www.sciencedirect.com/science/article/pii/S0031320323004260.

[9] Chen G, Zhou L, Zhang J, Yin X, Cui L, Dai Y. ESKNet: An enhanced adaptive selection kernel convolution for ultrasound breast tumors segmentation. Expert Syst Appl 2024;246:123265. http://dx.doi.org/10.1016/j.eswa.2024.123265, URL https://www.sciencedirect.com/science/article/pii/S0957417424001301.

[10] Chen G, Dai Y, Zhang J. C-Net: Cascaded convolutional neural network with global guidance and refinement residuals for breast ultrasound images segmentation. Comput Methods Programs Biomed 2022;225:107086. http://dx.doi.org/10.1016/j.cmpb.2022.107086, URL https://www.sciencedirect.com/science/article/pii/S0169260722004679.

[11] Zou Z, Chen K, Shi Z, Guo Y, Ye J. Object detection in 20 years: A survey. Proc IEEE 2023;111(3):257–76.

[12] Yap MH, Goyal M, Osman F, Martí R, Denton E, Juette A, Zwiggelaar R. Breast Ultrasound Region of interest detection and lesion localisation. Artif Intell Med 2020;107:101880. http://dx.doi.org/10.1016/j.artmed.2020.101880.

[13] Daoud MI, Al-Ali A, Alazrai R, Al-Najar MS, Alsaify BA, Ali MZ, Alouneh S. An Edge-Based Selection Method for Improving Regions-of-Interest Localizations Obtained Using Multiple Deep Learning Object-Detection Models in Breast Ultrasound Images. Sensors (Basel, Switzerland) 2022;22(18). http://dx.doi.org/10.3390/s22186721, URL https://pubmed.ncbi.nlm.nih.gov/36146070 arXiv:36146070.

[14] Ren S. Faster R-cnn: Towards real-time object detection with region proposal networks. 2015, arXiv preprint arXiv:1506.01497 arXiv:1506.01497.

[15] Marmé F, Krieghoff-Henning E, Gerber B, Schmitt M, Zahm D-M, Bauerschlag D, Forstbauer H, Hildebrandt G, Ataseven B, Brodkorb T, Denkert C, Stachs A, Krug D, Heil J, Golatta M, Kühn T, Nekljudova V, Gaiser T, Schönmehl R, Brochhausen C, Loibl S, Reimer T, Brinker TJ. Deep learning to predict breast cancer sentinel lymph node status on INSEMA histological images. Eur J Cancer (Oxford, Engl : 1990) 2023;195:113390. http://dx.doi.org/10.1016/j.ejca.2023.113390.

[16] Wetstein SC, de Jong VMT, Stathonikos N, Opdam M, Dackus GMHE, Pluim JPW, van Diest PJ, Veta M. Deep learning-based breast cancer grading and survival analysis on whole-slide histopathology images. Sci Rep 2022;12(1):15102. http://dx.doi.org/10.1038/s41598-022-19112-9.

[17] Balasubramanian AA, Al-Heejawi SMA, Singh A, Breggia A, Ahmad B, Christman R, Ryan ST, Amal S. Ensemble Deep Learning-Based Image Classification for Breast Cancer Subtype and Invasiveness Diagnosis from Whole Slide Image Histopathology. Cancers 2024;16(12). http://dx.doi.org/10.3390/cancers16122222.

[18] Hameed Z, Zahia S, Garcia-Zapirain B, Javier Aguirre J, María Vanegas A. Breast Cancer Histopathology Image Classification Using an Ensemble of Deep Learning Models. Sensors (Basel, Switzerland) 2020;20(16). http://dx.doi.org/10.3390/s20164373.

[19] Shao Y, Hashemi HS, Gordon P, Warren L, Wang J, Rohling R, Salcudean S. Breast Cancer Detection Using Multimodal Time Series Features From Ultrasound Shear Wave Absolute Vibro-Elastography. IEEE J Biomed Health Informat. 2022;26(2):704–14. http://dx.doi.org/10.1109/JBHI.2021.3103676.

[20] Chang JM, Won J-K, Lee K-B, Park IA, Yi A, Moon WK. Comparison of shear-wave and strain ultrasound elastography in the differentiation of benign and malignant breast lesions. AJR Am J Roentgenol 2013;201(2):W347–56. http://dx.doi.org/10.2214/AJR.12.10416.

[21] Qian X, Pei J, Zheng H, Xie X, Yan L, Zhang H, Han C, Gao X, Zhang H, Zheng W, Sun Q, Lu L, Shung KK. Prospective assessment of breast cancer risk from multimodal multiview ultrasound images via clinically applicable deep learning. Nat Biomed Eng 2021;5(6):522–32. http://dx.doi.org/10.1038/s41551-021-00711-2.

[22] Núñez C. Blood-based protein biomarkers in breast cancer. Clin Chim Acta; Int J Clin Chem 2019;490:113–27. http://dx.doi.org/10.1016/j.cca.2018.12.028.

[23] Loke SY, Lee ASG. The future of blood-based biomarkers for the early detection of breast cancer. Eur J Cancer (Oxford, Engl : 1990) 2018;92:54–68. http://dx.doi.org/10.1016/j.ejca.2017.12.025.

[24] Wang Y, An R, Yu H, Dai Y, Lou L, Quan S, Chen R, Ding Y, Zhao H, Wu X, Liu Z, Wang Q, Gao Y, Xie X, Zhang J. Largescale multicenter study of a serum metabolite biomarker panel for the diagnosis of breast cancer. IScience 2024;27(7):110345. http://dx.doi.org/10.1016/j.isci.2024.110345.

[25] Lokaj B, Durand de Gevigney V, Djema D-A, Zaghir J, Goldman J-P, Bjelogrlic M, Turbé H, Kinkel K, Lovis C, Schmid J. Multimodal deep learning fusion of ultrafast-DCE MRI and clinical information for breast lesion classification. Comput Biol Med 2025;188:109721. http://dx.doi.org/10.1016/j.compbiomed.2025.109721, URL https://www.sciencedirect.com/science/article/pii/S001048252500071X.

[26] Li M-D, Li W, Lin M-X, Lin X-X, Hu H-T, Wang Y-C, Ruan S-M, Huang Z-R, Lu R-F, Li L, Kuang M, Lu M-D, Chen L-D, Wang W, Huang Q-h. Systematic comparison of deep-learning based fusion strategies for multi-modal ultrasound in diagnosis of liver cancer. Neurocomputing 2024;603:128257. http://dx.doi.org/10.1016/j.neucom.2024.128257, URL https://www.sciencedirect.com/science/article/pii/S0925231224010282.

[27] Du J, Zhang Z, Tao W. MultiBCD: A multimodal model that simulates the human diagnostic process for automated breast cancer detection. Biomed Signal Process Control 2025;104:107410. http://dx.doi.org/10.1016/j.bspc.2024.107410, URL https://www.sciencedirect.com/science/article/pii/S174680942401468X.

[28] Ranjbarzadeh R, Dorosti S, Jafarzadeh Ghoushchi S, Caputo A, Tirkolaee EB, Ali SS, Arshadi Z, Bendechache M. Breast tumor localization and segmentation using machine learning techniques: Overview of datasets, findings, and methods. Comput Biol Med 2023;152:106443. http://dx.doi.org/10.1016/j.compbiomed.2022.106443, URL https://www.sciencedirect.com/science/article/pii/S0010482522011519.

[29] An Zichen LF. Advancements in the Application of Convolutional Neural Networks in Ultrasound Imaging for Breast Cancer Diagnosis and Treatment. Adv. Ultrasound Diagn. Ther. 2025;9(1):21–31, URL https://www.journaladvancedultrasound.com.

[30] Huang S, Lu Z, Cun X, Yu Y, Zhou X, Shen X. DEIM: DETR with Improved Matching for Fast Convergence. arXiv 2024. 2024, arXiv preprint arXiv:2412.04234 arXiv:2412.04234.

[31] Peng Y, Li H, Wu P, Zhang Y, Sun X, Wu F. D-FINE: Redefine regression Task in DETRs as Fine-grained distribution refinement. 2024, arXiv preprint arXiv:2410.13842 arXiv:2410.13842.

[32] Tian Y, Ye Q, Doermann D. Yolov12: Attention-centric real-time object detectors. 2025, arXiv preprint arXiv:2502.12524. arXiv:2502.12524.

[33] Lin T-Y, Goyal P, Girshick R, He K, Dollár P. Focal loss for dense object detection. In: Proceedings of the IEEE international conference on computer vision. 2017, p. 2980–8.

[34] Sohan M, Sai Ram T, Rami Reddy CV. A review on Yolov8 and its advancements. In: International conference on data intelligence and cognitive informatics. Springer; 2024, p. 529–45.

[35] Zhao Y, Lv W, Xu S, Wei J, Wang G, Dang Q, Liu Y, Chen J. Detrs beat yolos on real-time object detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024, p. 16965–74.

[36] Lv W, Zhao Y, Chang Q, Huang K, Wang G, Liu Y. Rt-Detrv2: Improved baseline with bag-of-freebies for real-time detection transformer. 2024, arXiv preprint arXiv:2407.17140 arXiv:2407.17140.

[37] Sohn K, Zhang Z, Li C-L, Zhang H, Lee C-Y, Pfister T. A simple semi-supervised learning framework for object detection. 2020, arXiv preprint arXiv:2005.04757 arXiv:2005.04757.

[38] Al-Dhabyani W, Gomaa M, Khaled H, Fahmy A. Dataset of breast ultrasound images. Data Brief 2020;28:104863.

[39] Ardakani AA, Mohammadi A, Mirza-Aghazadeh-Attari M, Acharya UR. An open-access breast lesion ultrasound image database: Applicable in artificial intelligence studies. Comput Biol Med 2023;152:106438.